

The Economy

Seminar 10: Trade

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Courtesy of
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- Increased integration of national economies into a global economy.
- Increased flows of goods, services, capital, and people.
- Two groups of globalization:
 - Trade globalization (goods and services)
 - Financial globalization (financial assets)
- Trade globalization often measured by the trade openness index, or simply the export-GDP ratio:

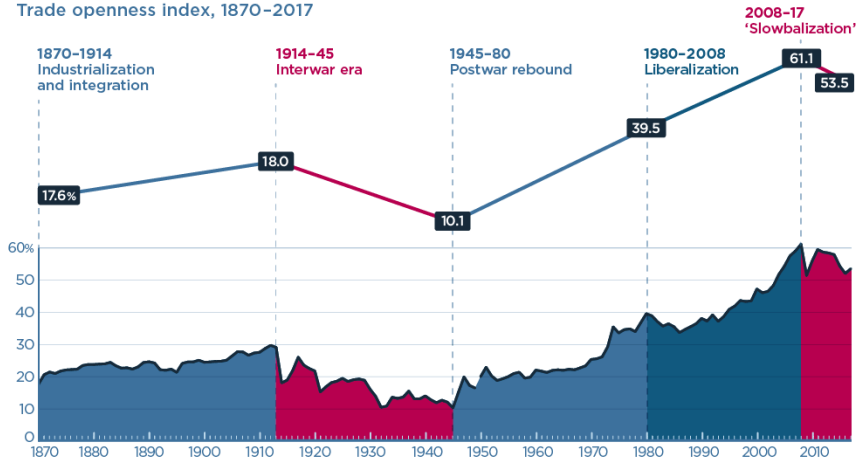
$$\text{trade openness index} = \frac{\text{exports} + \text{imports}}{\text{GDP}}$$

- Financial globalization measured by the financial openness index:

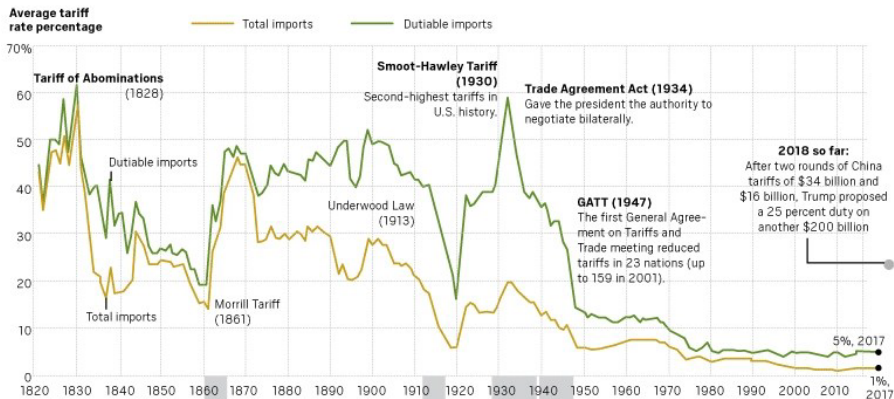
$$\text{financial openness index} = \frac{\begin{array}{c} \text{value of fin.} \\ \text{assets abroad} \end{array} + \begin{array}{c} \text{value of domestic assets} \\ \text{owned by foreigners} \end{array}}{\text{GDP}}$$

Globalization is in retreat for the first time since the Second World War

Trade openness index, 1870–2017



Trade openness: Lower trade barriers



Source: U.S. Department of Commerce, Bureau of the Census, U.S. International Trade Commission, Office of the United States Trade Representative

Civil War,
1861-1865

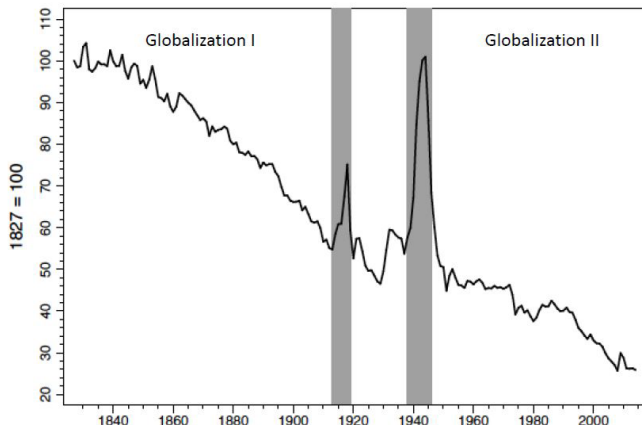
World War I,
1914-1918

Great
Depression,
1929-1939

World War II,
1939-1945

Trade openness: Trade costs

- Evolution of average world international relative trade costs: 1827-2014
 - trade costs: transportation, policies, communication, exchange rate volatility



Note: The variation of average trade costs is estimated on two-year windows, using balanced samples. The coefficients are then chained to obtain an index that covers the entire period.

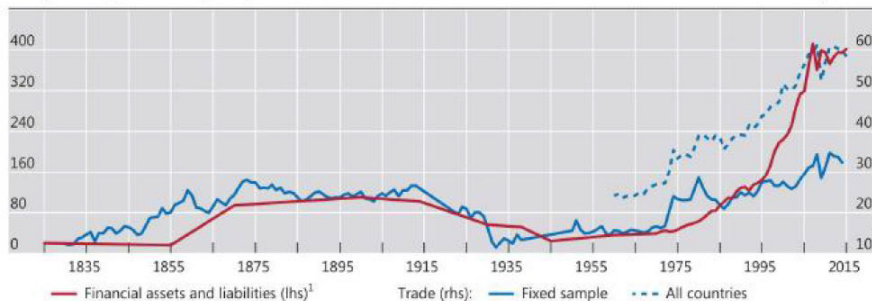
Source: Fouquot & Hugin 2016

Trade openness: Trade costs

- financial and trade globalization were traded together until 1990s but **decoupled** thereafter
 - driven by increased pervasiveness of the internet

As a percentage of country sample GDP

Graph VI.2

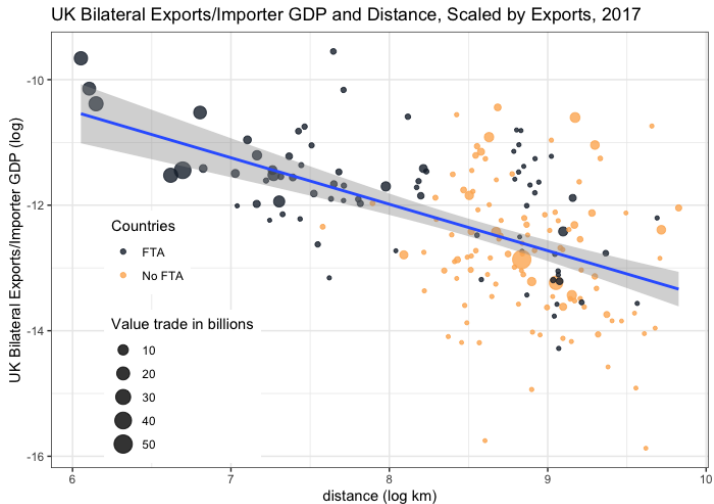


¹ Prior to 1970, calculated as external financial assets multiplied by two.

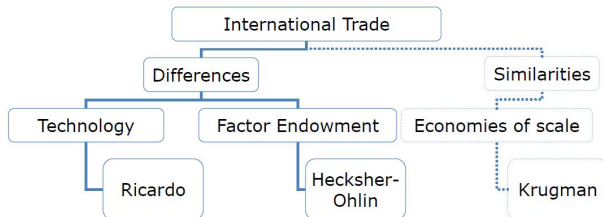
Sources: Federico and Tena-Junguito (2017); Lane and Milesi-Ferretti (2017); Obstfeld and Taylor (2004); Federal Reserve flow of funds accounts; IMF, *Balance of Payments Statistics*; World Bank; US Department of the Treasury; McKinsey Global Institute analysis; BIS calculations.

- the motor of globalization is trade, but the nature of the traded goods and the relative trade balances (between exporters and importers) has evolved
 - trade has moved away from agricultural and fuel products (primary sector: 60% → 30%) to manufacturing (secondary sector 40% → 70%), with services a comparatively small part
 - trade was originally among Western countries but has since shifted to include primarily emerging markets: in the BRIC markets, trade volume has grown exponentially, especially since they integrated the WTO
- but trade is not as "globalized" as it might first seem for those of us who live in Western Europe:
 - trade is often restricted to a limited set of goods and services
 - trade and role of globalization depends heavily on geographical proximity: volume of trade decreases in the geographical distance between trading countries = this is part of a theory known as gravity
- what explains the trade between countries?

$$\text{Trade volume} = \frac{1}{\alpha} \text{Distance}$$



Explaining trade



- **Ricardo:** countries specialize in good where they have the greatest relative advantage in TFP
- **Heckscher-Ohlin:** differences in comparative advantage comes from differences in factor abundance and intensity
- **Krugman:** countries tend to export those kind of products for which they have increasing economies of scale

Explaining trade

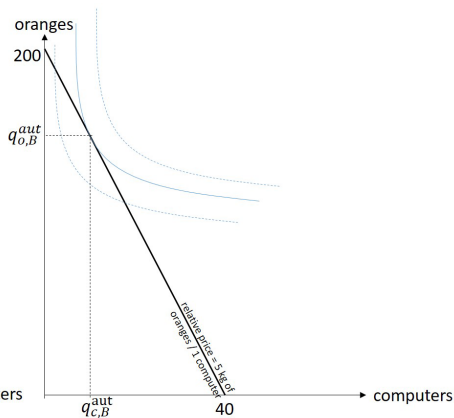
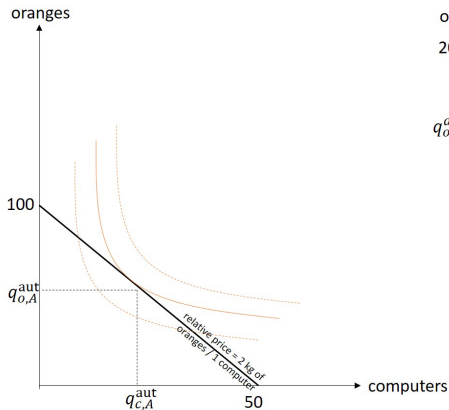
- with gains of specialization, countries can then trade
- economies of scale and of agglomeration = artificial reasons to specialize (cf. Krugman)
 - economies of scale: doubling all inputs to a production process more than doubles the output ($F(mK, mL) < mF(K, L)$) for a given firm
 - economies of agglomeration: firms that work in the same area share the same innovation and productivity gains, for a given region or country
- difference in technology or availability of production factors between regions due to historical or geographical endowments = natural reasons to specialize (cf. Ricardo or Heckscher-Ohlin)

Ricardo: Comparative Advantage

- Ricardo's theory explains trade by comparative productivity, not absolute productivity → explains how all countries can (and do) benefit from trade by specializing.
 - absolute advantage: the ability to produce one unit of good with fewer number of units than in another country
 - comparative advantage: the tradeoff between producing one good relative to another is less costly (the opportunity cost of producing one good is lower)
- Example with two countries (A and B):
 - Country A: 1 kilo of orange costs 2 labor units, a computer costs 4 labor units.
 - Country B: 1 kilo of orange costs 5 labor units, a computer costs 25 labor units.
 - the original endowment in A is 200 labor units and in B is 1000 labor units
- Comparative advantage allows countries to specialize and trade, thereby gaining more than in autarky.

Ricardo: Autarky

- on the left: country A; on the right: country B



Ricardo: Autarky

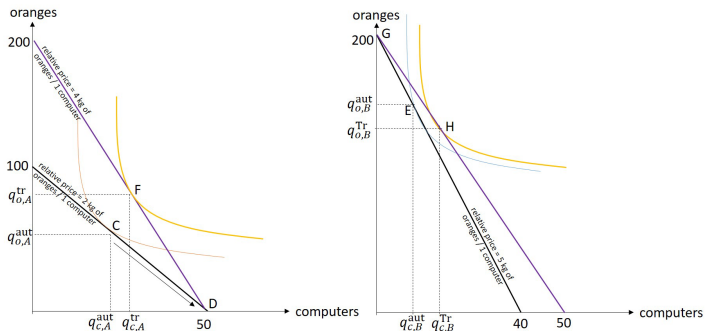
	A	B		A	B
oranges	2	5	opp. cost of oranges	$2/4 = 0.5$	$5/25 = 0.2$
computer	4	25	opp. cost of computer	$4/2 = 2$	$25/5 = 5$

- **A has an absolute advantage:** it needs less units of labor to produce the same output as B for both goods ($2 < 5$ and $4 < 25$)
- A has a **comparative advantage** in producing computers (its opportunity cost for producing computers - in terms of oranges - is lower: $2 < 5$)
- B has a **comparative advantage** in producing oranges (its opportunity cost for producing oranges - in terms of computers - is lower: $0.2 < 0.5$)
- without trade = in autarky, the relative price of computers is low in A (2) and high in B (5)
- trade allows countries with a higher relative price to buy from countries with a lower relative price

Ricardo: Trade

- without trade, the relative price of computers is 2 in A and 5 in B; and the relative price of oranges is 0.2 in B and 0.5 in A
- so if countries are allowed to trade: A will export computers – where it has a lower relative price than B and import oranges – where it has a higher relative price than B
 - then the relative price of computers after trade will converge to somewhere between 2 and 5 (i.e. it falls in B and increases in A) - let's say it is 4
 - then the relative price of oranges after trade will converge also to somewhere between 0.2 and 0.5 (i.e. it falls in A and increases in B) - let's say it is 0.25
 - the exact post-trade relative prices are not easy to determine: they depend on institutional factors, bargaining powers between countries, etc.
- trade makes both countries better off
 - previously, country A contented with C but country A can specialize in computers along the original feasible production frontier, producing at D (50 computers)
→ with trade, the relative price of computers increase from 2 to 4 in country A, making it possible to now consume at F, which is strictly better than C (trading off 50 computers with now 200 oranges instead of 100)
 - currently, country B contented with E but country B can specialize in oranges along their original feasible production frontier, producing at G (200 oranges)
→ with trade, the relative price of oranges increases from 0.2 to 0.25 in country B, making it possible to consumer at H (trading off 200 computers with now 50 oranges instead of 40) along a new feasible consumption frontier

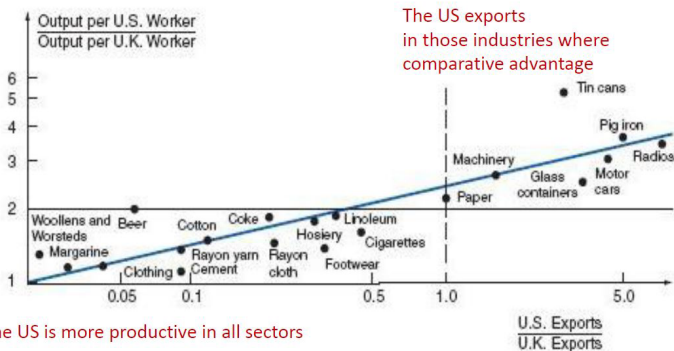
Ricardo: Trade



- how the gains of trade are distributed depends fully on how far the post-trade relative price is to each country's relative price in autarky
 - the closer the post-trade relative price to the autarky price in a country, the lower the gain in trade for that good
- specialization in linear production frontiers means any specialization is full (produce one good only)
 - but specialization in concave production frontiers means there is specialization but it may not be full = the diminishing MRT between goods means that the gain in welfare that might be possible might be outweighed by the amount of the other good we'd need to give up at that MRT

Trade: Absolute vs. Comparative advantage

- it is possible for a country to have an absolute disadvantage in all the industries vs. another a country and have comparative advantages in only some of them



Relative Labour Productivities and Comparative Advantage – United States and United Kingdom in 1937. The figure shows a positive relationship between labour productivity and export shares for 20 industries in the United States and the United Kingdom, thus confirming the Ricardian trade model.

Source: Adapted from G.D.A. MacDougall, "British and American Exports: A Study Suggested by the Theory of Comparative Costs" *Economic Journal*, December 1951, p.703

PS10 Exercise 1: Comparative advantage

We imagine a world with just two countries, Moria and Arrakis. Each country has 240 workers. The following table shows the number of workers needed to produce 1kg of ore and 1kg of spices over a given period in each country.

	Moria	Arrakis
Ores	2 workers	3 workers
Spices	3 workers	2 workers

- 1 Calculate the relative price of spices in each country, when they are self-sufficient. This is the value of spices expressed in terms of quantity of ore (to produce 1kg of spices, how many kg of ore must be given up?).
- 2 In which country are spices the "cheapest" in terms of ore? What does this tell you about the structure of comparative advantage?

PS10 Exercise 1: Comparative advantage

- 3 Calculate the maximum quantity of ore and spice that each country could produce, if it had devoted its entire workforce to one of the two sectors. Plot these figures on two graphs (one for each country), with spice quantities on the horizontal axis and ore quantities on the vertical axis. Draw the production possibilities frontier (possible combinations of ore and silver that each country can produce).
- 4 It is assumed that, in the absence of trade, Moria consumes 90kg of ore and 20kg of spice, and Arrakis consumes 20kg of ore and 90kg of spice. Place these points on your graph.

PS10 Exercise 1: Comparative advantage

- ⑤ Let's suppose that these countries start trading. At what relative price does it become worthwhile for Moria to procure spices through trade rather than producing it themselves? At what relative price does it become worthwhile for Arrakis to sell spices on the world market rather than producing it themselves? Using this, find the possible range of values for the relative world price of spices.
- ⑥ If the relative world price of spices is 1, how will countries specialize? Draw the consumption possibilities frontier made possible by trade in each country. Using these graphs, show that trade is mutually beneficial.

- **PS11 Exercise 1** due in class, no text!
- Quiz 6 next week.
- Final exam on the 6th of May.
 - Lectures 6-12!
- Mock exams on Moodle next week.